An Agentic Framework Using Slime Mold Intelligence for Transportation Network Optimization

Jitendra[1], Nimal S[2], Roshan Paul Patrick[3], Dr. R. Anandan[4]

[4]Professor, [1,2,3]UG Scholars

Department of Artificial Intelligence and Machine Learning,
Vels Institute of Technology
Pallavaram, India

Email id : jitendrasundesha2006@gmail.com , s.nimal234234@gmail.com , excallibur493@gmail.com , anandan.se@vistas.ac.in

Abstract

Designing transportation networks that connect many stations with limited infrastructure, while staying reliable when individual links fail, is a central challenge in smart urban mobility. Biological systems offer an alternative to centralized optimization: the slime mold *Physarum polycephalum* self-organizes efficient, fault-tolerant transport networks between food sources. This paper presents an agentic framework that implements the agent-based, raster-based Jones model, in which a population of simple autonomous agents senses, deposits, and follows a diffusing chemoattractant trail, with stations acting as nutrient stimuli. Because the process is stochastic, the simulation is repeated over multiple independent runs. A station-to-station graph is extracted from each trail field through thresholding, skeletonization, and station association, and the graphs are merged into a consensus network in which every link carries an edge frequency, the fraction of successful runs in which it appears. Links above a user-defined frequency threshold are treated as stable connections, while less frequent links reveal alternative candidate corridors. Developed using Python, NumPy, SciPy, NetworkX, and Streamlit, the system provides an interactive map for station entry and CSV import, adjustable model parameters, and exports in CSV, GeoJSON, and reproducible project formats for GIS tools. In a Chennai Metro case study, high-frequency links connected all stations, indicating the framework's potential as a transparent decision-support tool for early-stage network planning. Generated links are conceptual candidate alignments, not surveyed routes.

Keywords— Slime Mold Intelligence, *Physarum polycephalum*, Agentic Framework, Agent-Based Simulation, Transportation Network Optimization, Jones Model, Consensus Network, Geospatial Analysis.